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IT Management



EGIT503 Research Methodology

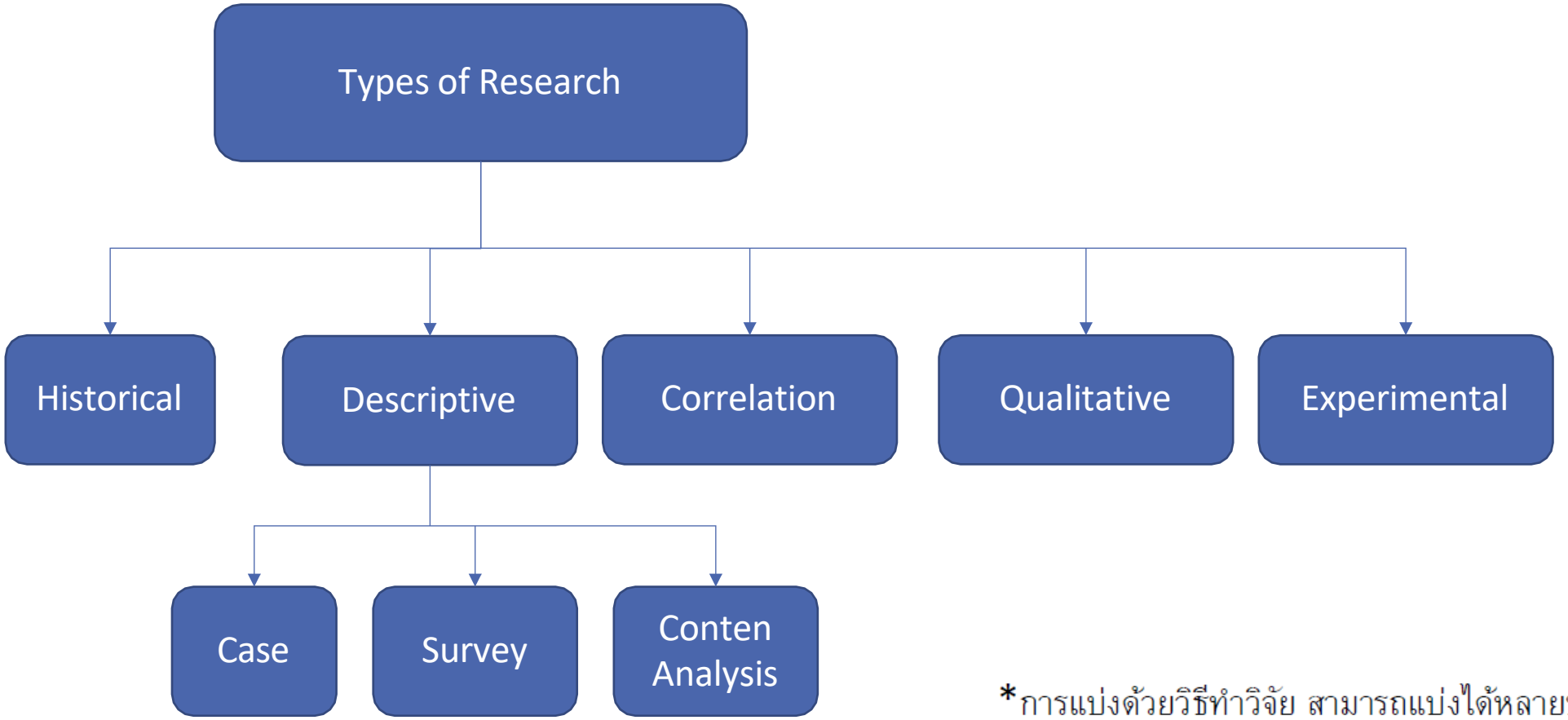
Lecture#2 Research Database and Quality of Research

ผศ.ดร.ยอด สุขะมงคล
อาจารย์ผู้รับผิดชอบรายวิชา

Criteria for Research Classification

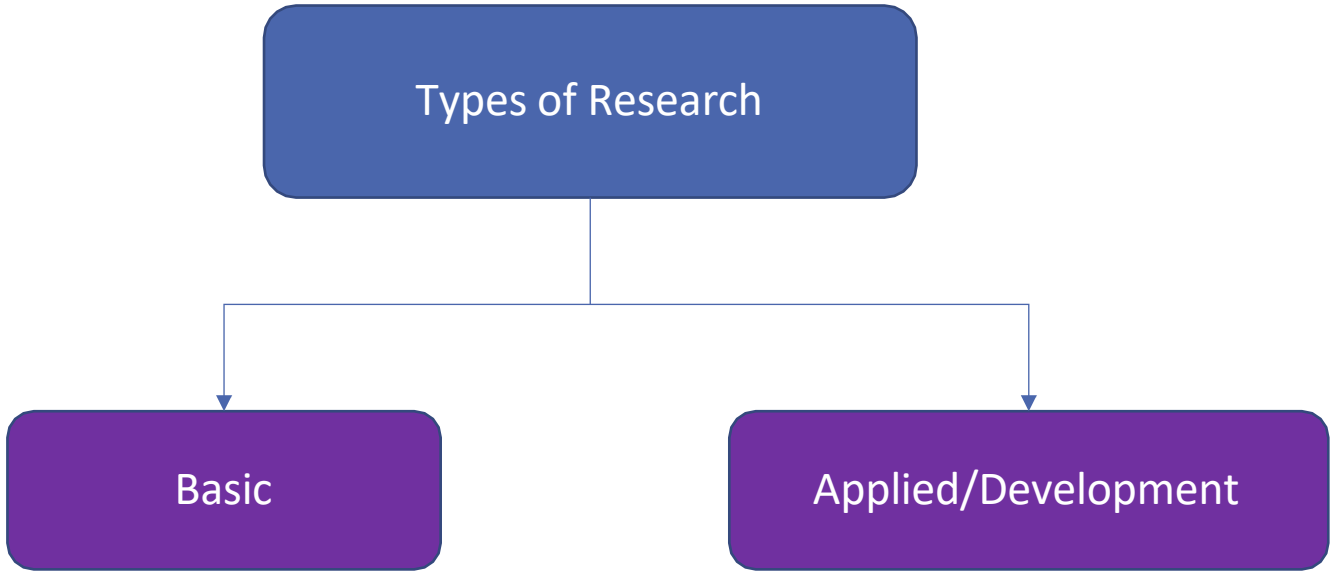
- การแบ่งประเภทของงานวิจัยมี 3 หลักเกณฑ์
  - แบ่งตามวิธีการวิจัย (methods)
  - แบ่งตามวัตถุประสงค์ (purposes)
  - แบ่งตามความสามารถในการทดลอง (ability to experiment)

Based on Methods

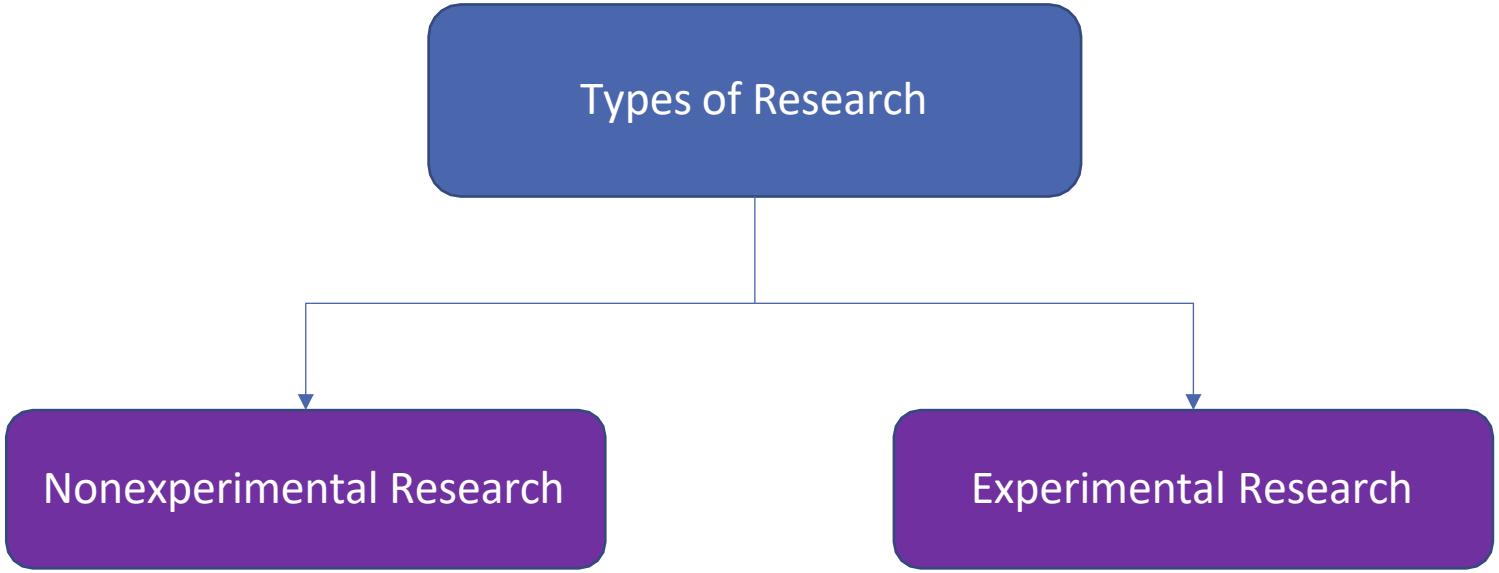


*การแบ่งด้วยวิธีทำวิจัย สามารถแบ่งได้หลายประเภทมาก ยังไม่สามารถสรุปได้แน่ชัดว่ามีกี่วิธี อะไรใดบ้าง ที่นำมาแสดงนี้เป็นเพียงตัวอย่างเท่านั้น แต่ละสาขาแม้จะมีวิธีที่เหมือนหรือคล้ายกันก็สามารถเรียกแตกต่างกันไปได้

Based on Purposes



Based on Ability to Experiment



Based on Methods and Ability to

	Types of Research					
	Nonexperimental				Experimental	
	Descriptive	Historical	Correlational	Qualitative	True Experimental	Quasi-experimental
Purpose	Describe the characteristics of an existing phenomenon	Relate events that have occurred in the past to current events	Examine the relationships between variables	To examine human behavior and the social, cultural, and political contexts within which it occurs	To test for true cause-and-effect relationships	To test for causal relationships without having full control
Time frame	Current	Past	Current or past (correlation) Future (prediction)	Current or past	Current	Current or past
Degree of control over factors or precision	None or low	None or low	Low to medium	Moderate to high	High	Moderate to high
Code words to look for in research articles	Describe Interview Review literature	Past Describe	Relationship Related to Associated with Predicts	Case study Evaluation Ethnography Historical Research Survey	Function of Cause of Comparison Between Effects of	Function of Cause of Comparison between Effects of
Example	A survey of dating practices of adolescent girls	An analysis of Freud's use of hypnosis as it relates to current psychotherapy practices	An investigation that focuses on the relationship between the number of hours of television watching and grade-point average	A case study analysis of the effectiveness of policies for educating all children	The effect of a preschool language program on the language skills of inner-city children	Gender differences in spatial and verbal abilities

Table 1.1 Summary of research methods covered in *exploring research*.

Find a topic

Narrow and deep topic	*Wide and shallow topic*
Advantages • More chance of pushing the boundary of knowledge • More exciting	Advantages • Realistic • Good training for industrial research
Disadvantages • Your “model” may be too abstract and unrealistic • It’s hard to choose the variable parameters	Disadvantages • Mostly boring, like a collection of honors theses • Unlikely to contribute a lot

Ex : New CNN, New Decision Tree Algorithm

Ex : Effect of AI, Social Emotion Detection

Find a good topic

Fundamental topic	*Applied topic*
Advantages • Your thesis will have a longer life • Your work can have more applications	Advantages • Easier problems • May help with getting a job in industry • Can contribute a lot to a relevant area
Disadvantages • It's hard to push the boundaries very far • Your "model" may be too abstract and unrealistic	Disadvantages • Your thesis can die young • Your employment prospects can be shortlived • Restricted applications

Ex : Maslow's Hierarchy of Need, Relativity Theory

Ex : 5G, IoT, Block Chain

Find a good topic

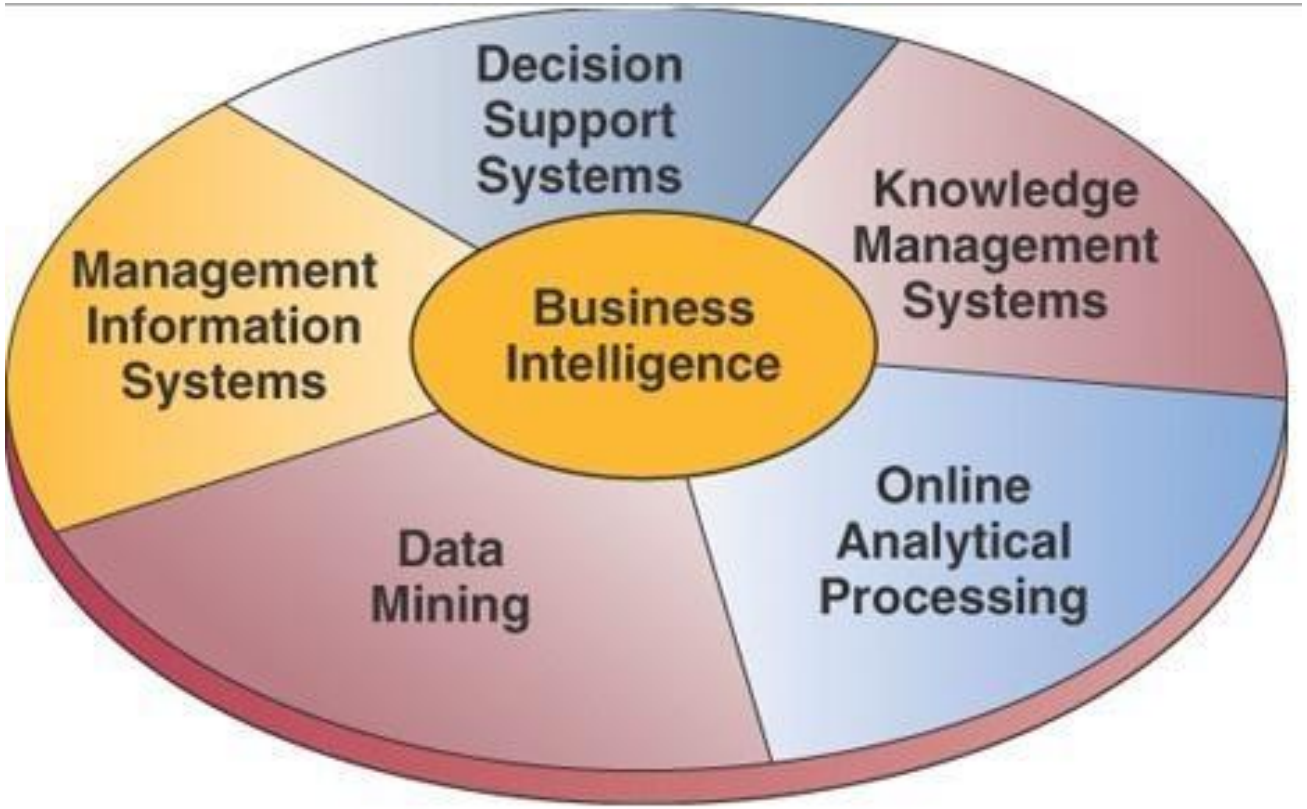
Classical topic	*Hot topic*
Advantages • You may win the lottery and solve a hard problem • Your thesis may have a long life • Better referees • Higher scientific quality	Advantages • Better immediate feedback • With good timing, you can get rich • Easier to publish • Vibrant community
Disadvantages • Can be frustrating • Immediate rewards can be small	Disadvantages • Your thesis can die young • Scientific quality can be low

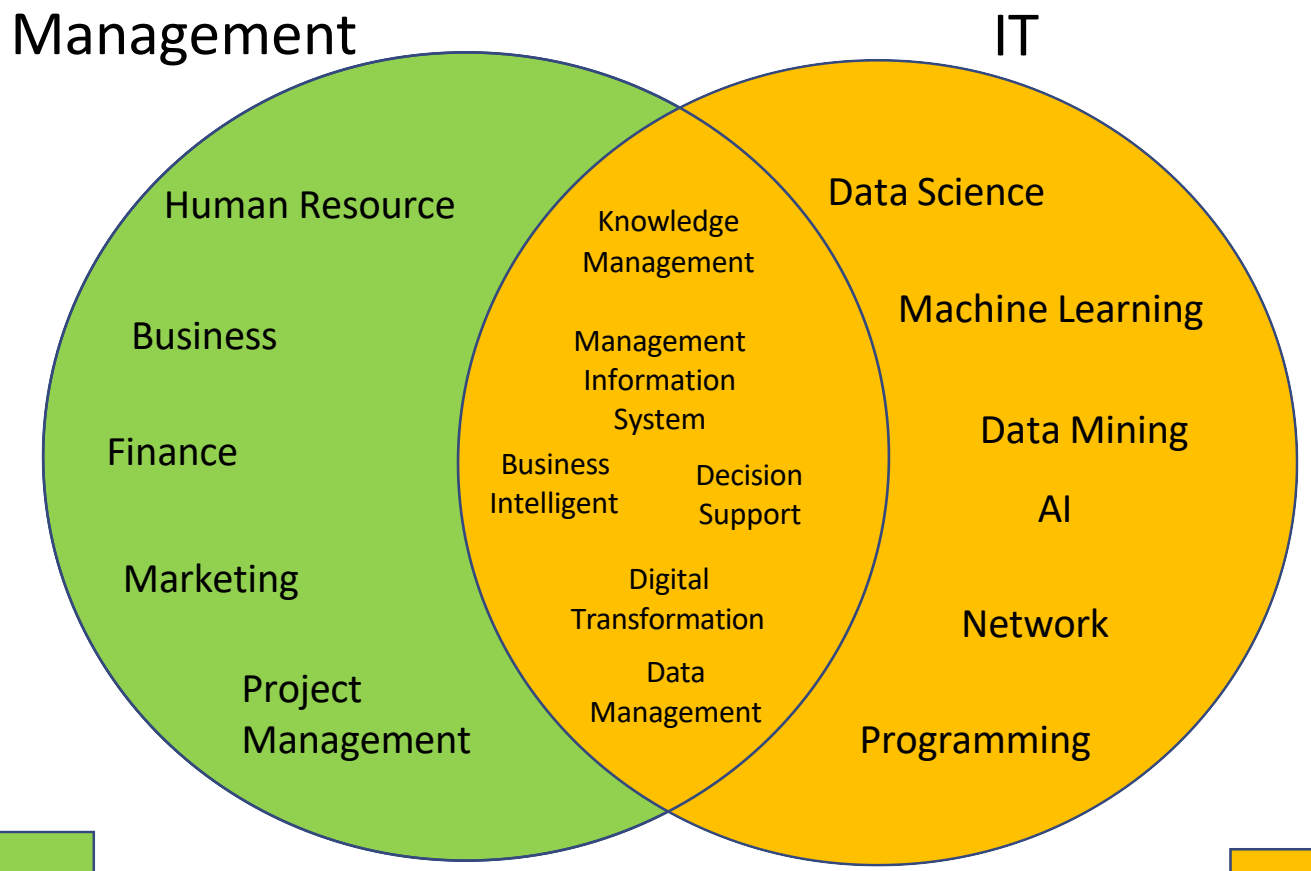
Ex : Black Hole Theory, Maslow's Hierarchy of Need, WPT

Ex : Block Chain, IoT, Fintech

Technology of Information System Management (TISM) Research Areas

TISM Research Area

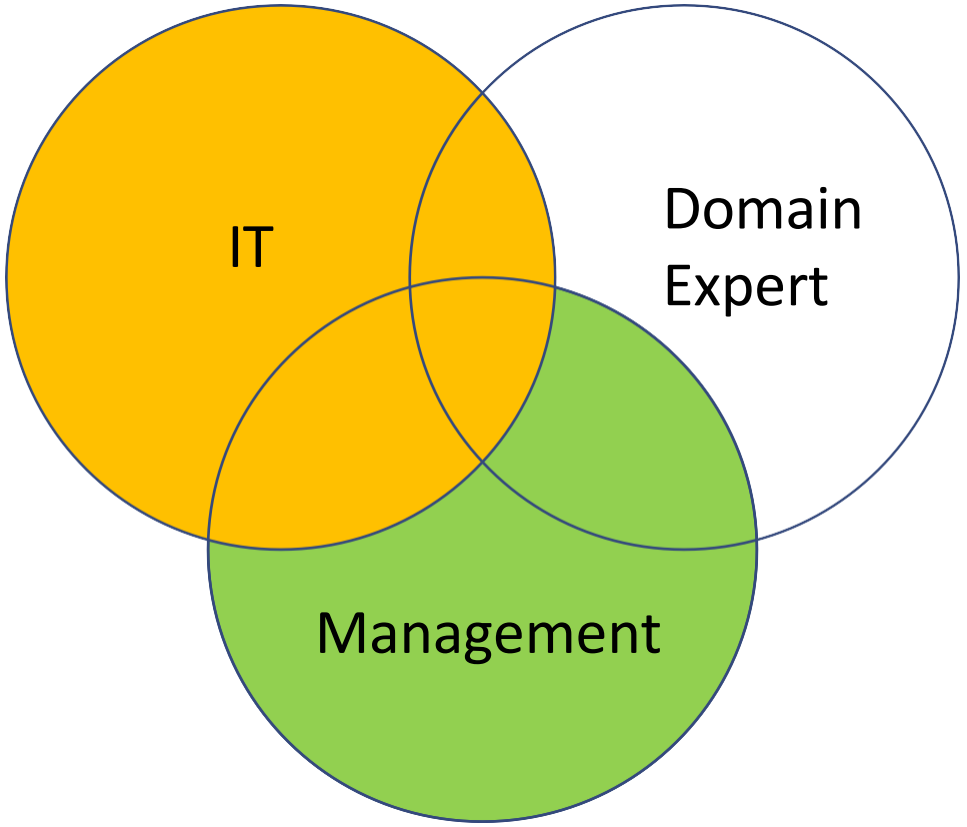




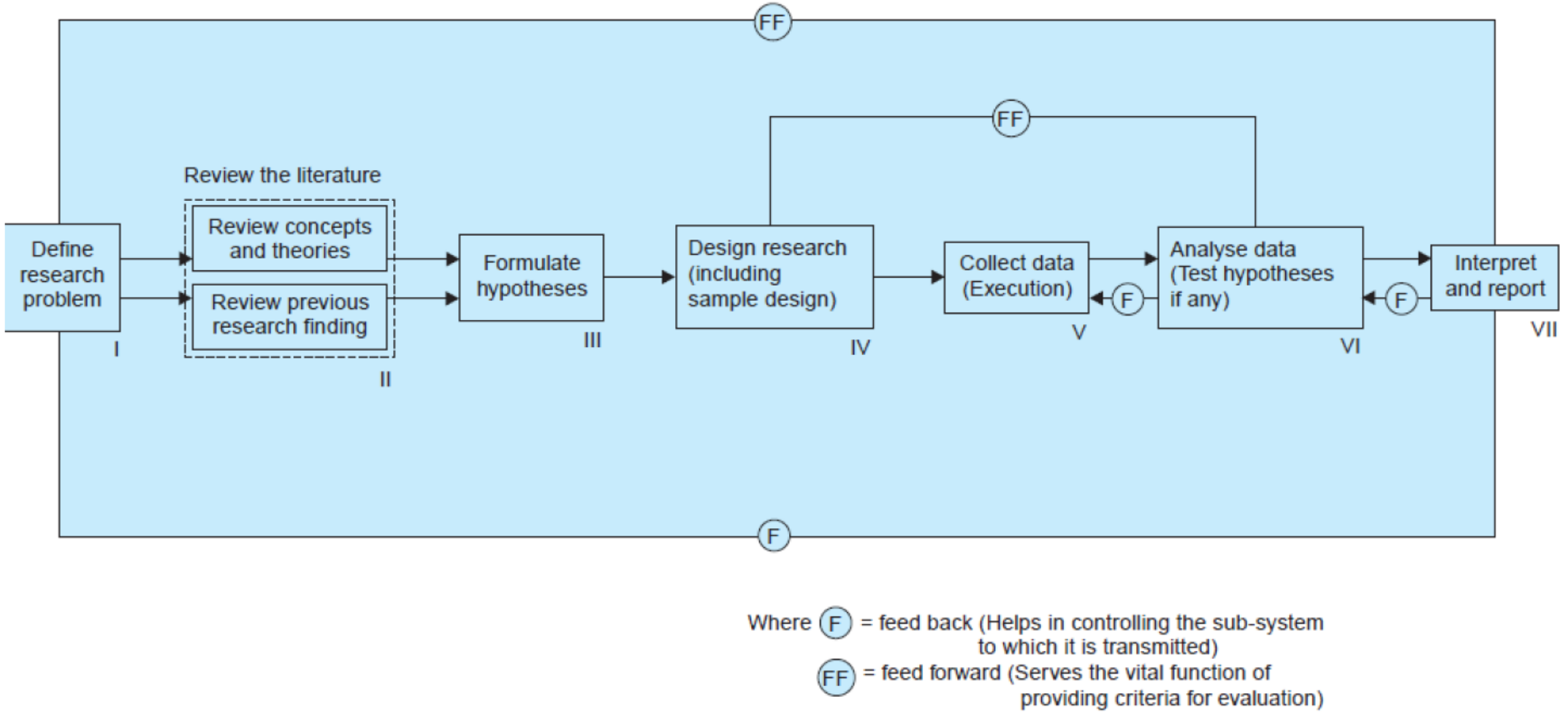
Method
Survey + Statistic, Conceptual and Empirical

Method
Data Analytic, Math Model, Simulation Model

TISM Research Area

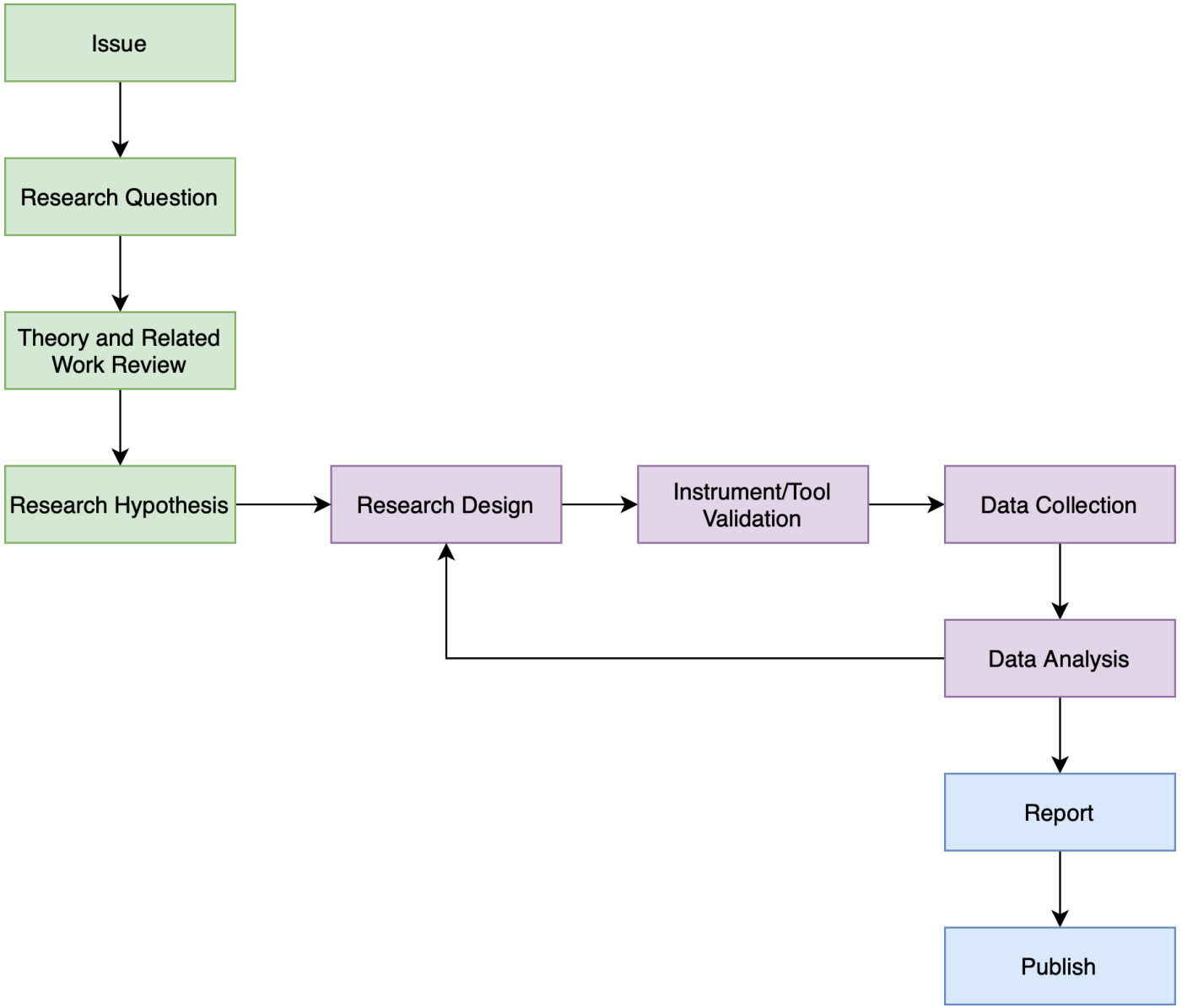


RESEARCH PROCESS IN FLOW CHART

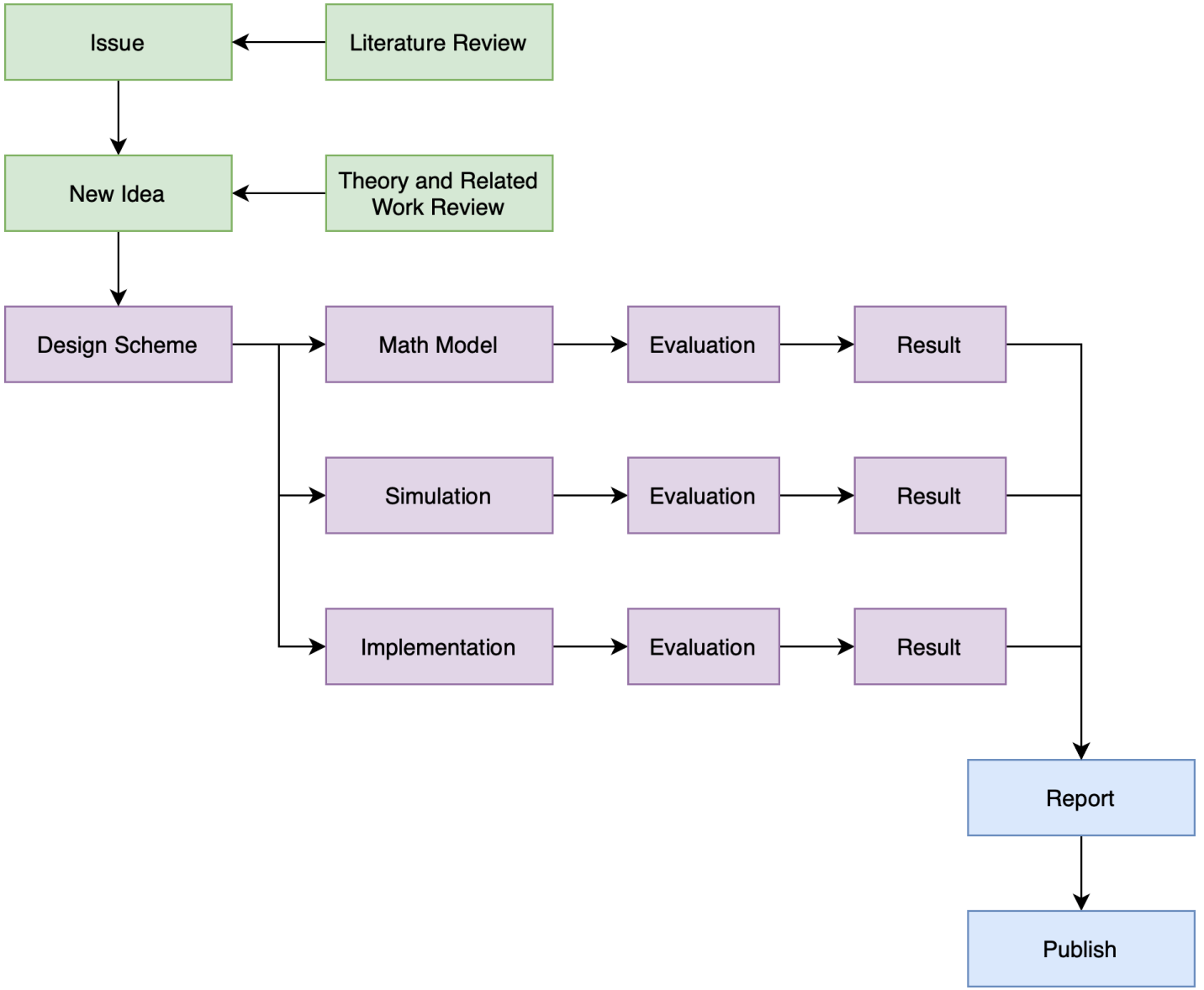


*C.R. Kothari , Research Methodology Methods and Techniques, 2004

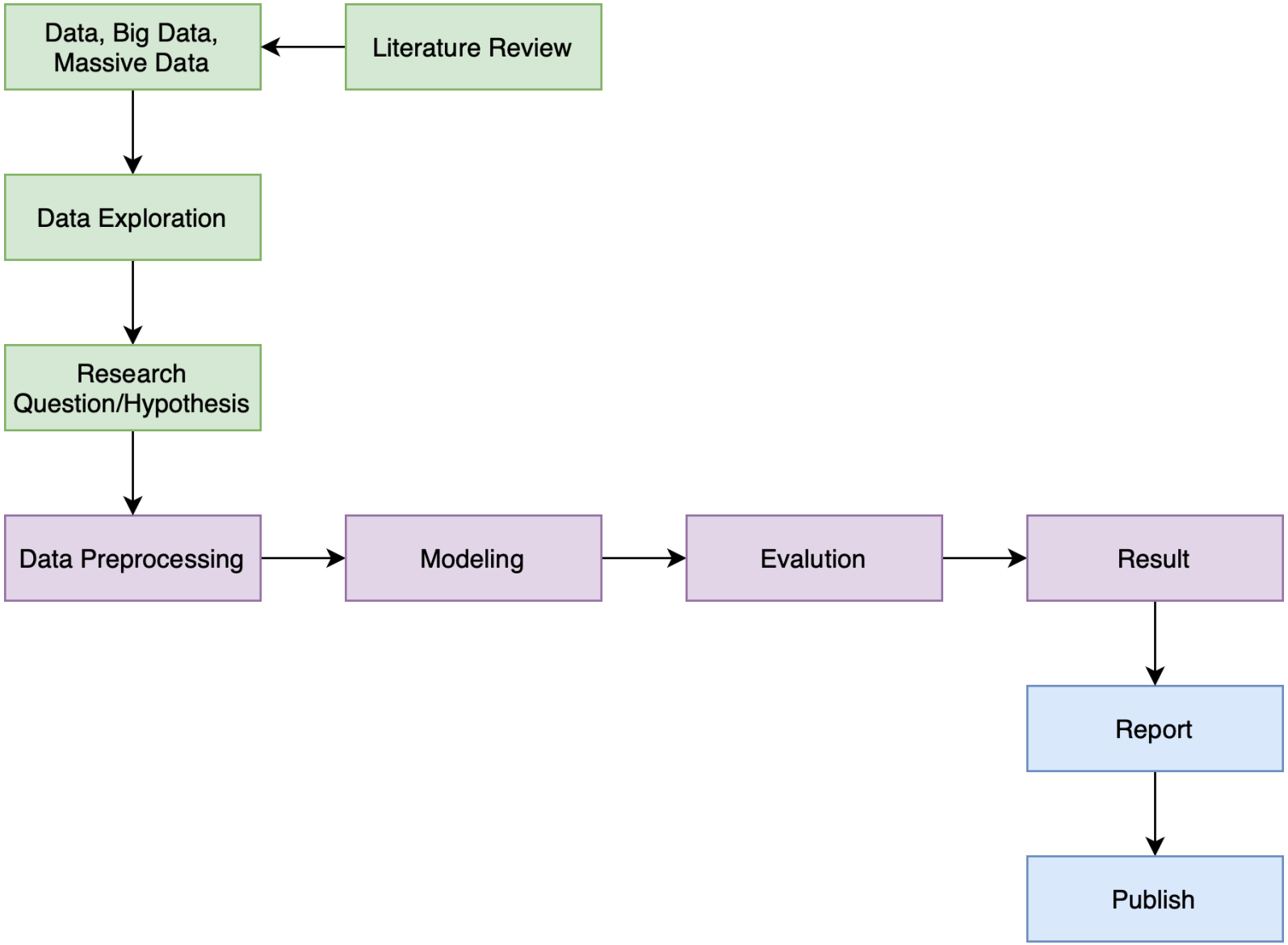
Social Science Research Process



Engineering Research Process



Data Science Research Process



Class Activity #1

1. Sunisa Nuntasing, Adisorn Leelesantitham and **Yod Sukamongkol**, "Predictions of Stem Skills in Human Resource Management Using Machine Learning", *The 2025 6th Technology Innovation Management and Engineering Science International Conference (TIMES-iCON2024)*, Dec 2025, Bangkok, Thailand
2. Piyawadee Pengna, Adisorn Leelasantitham and **Yod Sukamongkol,** "A Low-Code Platform of Carbon Credit Trading Using Blockchain Technology: A Case Study in Nakhon Si Thammarat Province", *The 2024 Technology Innovation Management and Engineering Science International Conference (TIMES-iCON2024)*, June 2024, Bangkok, Thailand
3. **Yod Sukamongkol**, Siriwarin Petcharat and Lek Lorsomrudee, "The Experimental Investigation on a Hybrid Electrical and Thermal Generation System with Thermoelectric and Heat-Pipe Solar Collector", *Management and Innovation Technology International Conference (MITiCON)*, December 2017, Ayutthaya, Thailand
4. **Yod Sukamongkol**, "Barriers of the Solar PV Rooftop Promoting in Thailand", *Management and Innovation Technology International Conference (MITiCON)*, Octorber 2016, Chonburi, Thailand
5. **Yod Sukamongkol**, "An Experimental Study of the Effect of Dust on Photovoltaic Power Generation Efficiency in a Tropical Climate", *Journal of Sustainable Energy & Environment*, 2015, Vol 6, Issue 2
6. Pornphimol Winyuchakrit and **Yod Sukamongkol**, "Analyses of CO_2 Mitigation in Thai Road Transportation: The Past to The Future", *Management and Innovation Technology International Conference (MITiCON)*, December, 2014, Chonburi, Thailand
7. **Yod Sukamongkol**, Supachart Chungpaibulpatana, and Bundit Limmeechokchai, "A Hybrid PV/Th Air Heating System for Regenerating Desiccant used in an Air Conditioning Room: Simulation and Experimental Validation", *World Renewable Energy Congress 2009 Asia*, Bangkok, Thailand, May 19-22, 2009.



Group Activity #1

- From the given articles, check the following information, one article per group.
  - Type of Topic?
  - Type of Research?
  - Draw the Research processes?



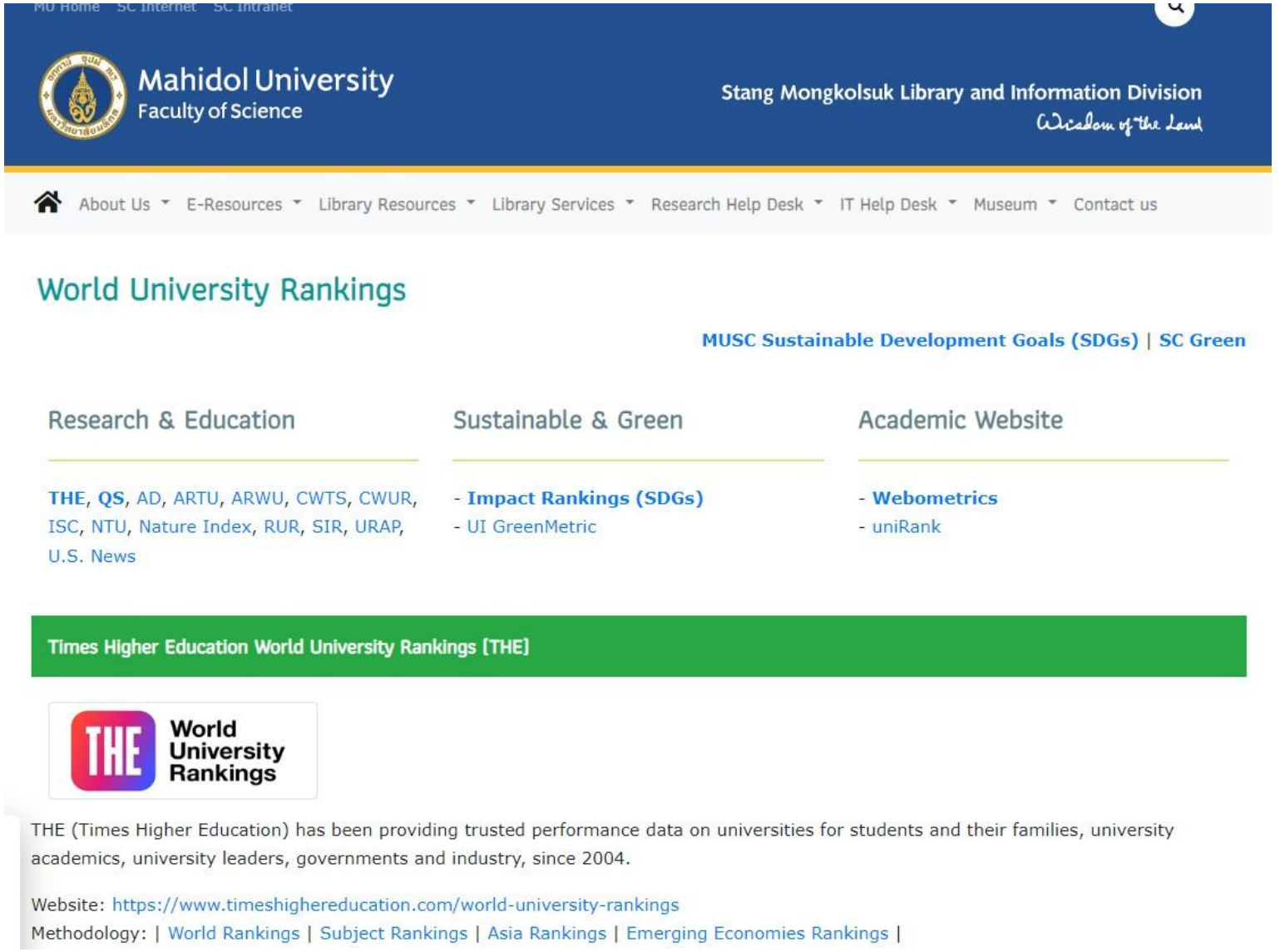
Research Database and Quality of Research

ข่าวงานวิจัย



https://www.youtube.com/watch?v=eNnxkL5h_do

World University Ranking





QS World University Rankings 2025

Time Higher Education

World University Ranking 2025

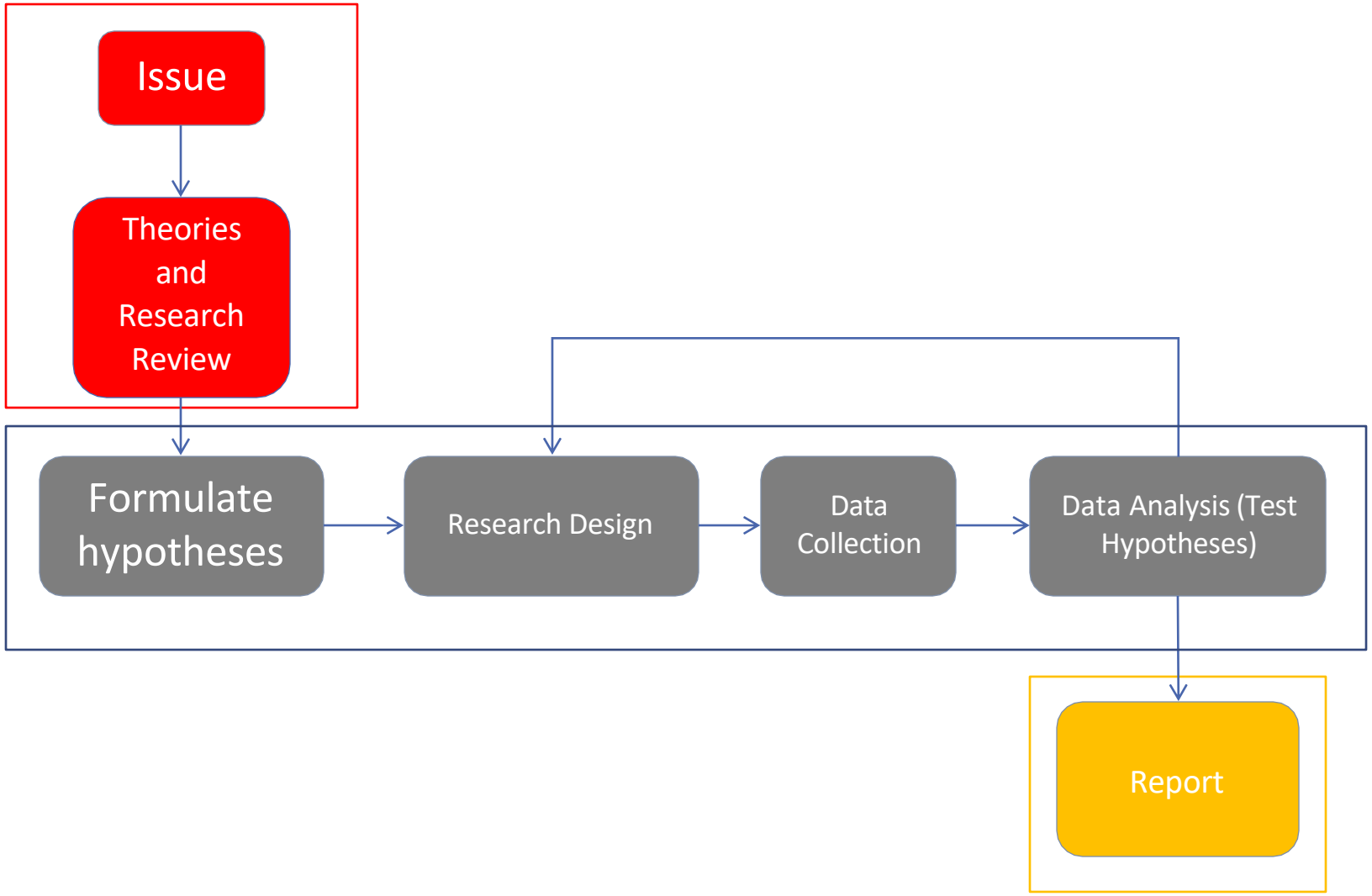


Thailand Rank	World Rank	University	Overall Score*	Teaching (29.5%)	Research Environment (29%)	Research Quality (30%)	Industry (4%)	International Outlook (7.5%)
1	601–800	Chulalongkorn University	42.6	36	32.4	53	81.6	45
2	601–800	Mahidol University	38.6	38.2	23.2	48.2	61.2	49
3	801–1,000	King Mongkut's University of Technology Thonburi	37.2	22.5	23.6	58	75.2	43.7
4	1,001–1,200	Chiang Mai University	33.1	28.6	19.4	46.5	52.2	39.9
5	1,001–1,200	Khon Kaen University	30.7	30.3	16.4	41.4	46.4	35.9
6	1,201–1,500	Walailak University	29.7	20.3	14.6	51.7	17.7	43.6
7	1,201–1,500	King Mongkut's University of Technology North Bangkok	29.6	16.8	13.2	55.8	46.9	29.3
8	1,201–1,500	Mae Fah Luang University	29.2	16	13.5	49.3	35.5	58.2
9	1,201–1,500	Prince of Songkla University	27.1	24.3	15.3	35.8	48.3	38.1
10	1,201–1,500	Maejo University	27.1	14.1	9.2	53.4	32.9	39.4
11	1,201–1,500	Thammasat University	26.6	27.9	20	28.1	32.8	37.9

ผลงานวิจัยส่งผลอะไรบ้าง?

- เผยแพร่ความรู้ใหม่
- สำเร็จการศึกษา
- เลื่อนตำแหน่งทางวิชาการ
- Ranking ของมหาวิทยาลัย

General Research Process



General Research Process

*C.R. Kothari , Research Methodology Methods and Techniques, 2004

Skills for Research



- Questioning Skill
- Searching Skill
- Reading Skill
- Listening Skill
- Watching Skill

- Questioning Skill
- Research Design Skill
- Related Area Skill
- Computer Skill
- Lab Skill
- Etc..

- Speaking Skill
- Writing Skill
- Presenting Skill
- Answering Skill

Knowledge Sources



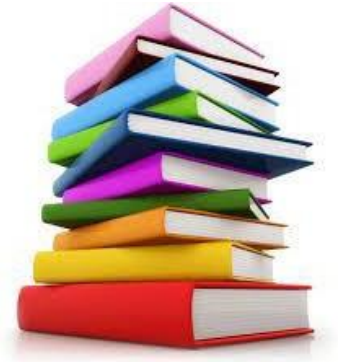
Quality Sources



Scholarly articles



Conference Proceedings



Books



Thesis and Dissertation

Knowledge Source

Google Scholar!!



Research Databases

How to search a Research Article

- Google ????
- Google Scholar !!!

What is Google Scholar?

- A free search engine for scholarly literature across various disciplines.
- Ideal for finding academic papers, theses, books, conference proceedings, and patents.

Tips for Effective Use

1.Search for Specific Phrases

1. Use quotation marks to find exact phrases.
2. Example: "artificial intelligence in healthcare"

2.Use Advanced Search

1. Access via menu (three horizontal lines) > **Advanced search**.
2. Refine by author, publication, date range, etc.

3.Check Citations

1. "Cited by" link shows the number of times a paper is cited.
2. Helps gauge the paper's influence.

4.Explore Related Articles

1. Click "Related articles" to find similar research.
2. Great for discovering additional relevant studies.

5.Save Articles for Later

1. Use the "Save" button to add articles to your personal library.
2. Organize with folders for easy access.

6. Export Bib for citation system

1. Export the list of paper to bibliography database for citation.

IEEE Explore Digital Library

- http://ieeexplore.ieee.org

Research Databases

- Research databases are **organized collections** of computerized information or data such as periodical articles, books, graphics and multimedia that can be searched to retrieve information.
- Databases can be general or subject oriented with bibliographic citations, abstracts, and or full text.
- The sources indexed may be written by scholars, professionals or generalists.

Research Databases

- Books (printed & e-book)
  - Web OPAC (MU Library Catalog)
  - EDS (EBSCO Discovery Service)
  - ScienceDirect (publisher = Elsevier)
  - SpringerLink (publisher = Springer)
  - and other publishers
- Articles in journal and magazine
  (currently almost in electronic form with printed achieved)
  - SCOPUS, EDS, PubMed, Web of Science
  - SciFinder, MathSciNet, ACS, RSC
- Theses (in PDF and printed achieved)

ELSEVIER

Springer

ScienceDirect

EBSCO Discovery Service

PubMed

Scopus

WEB OF SCIENCE™

Area of each database

- Multidisciplinary (content for all topics)
  - EDS (EBSCO Discovery Service)
  - ScienceDirect (publisher = Elsevier)
  - SpringerLink (publisher = Springer)
  - Wiley (publishers) etc…
- Medical and health science
  - PubMed (database)
  - Ovid, Kargeretc… (publisher)
- Chemistry
  - SciFinder (database), ACS, RSC (publisher)
- Mathematics
  - MathSciNet (database)

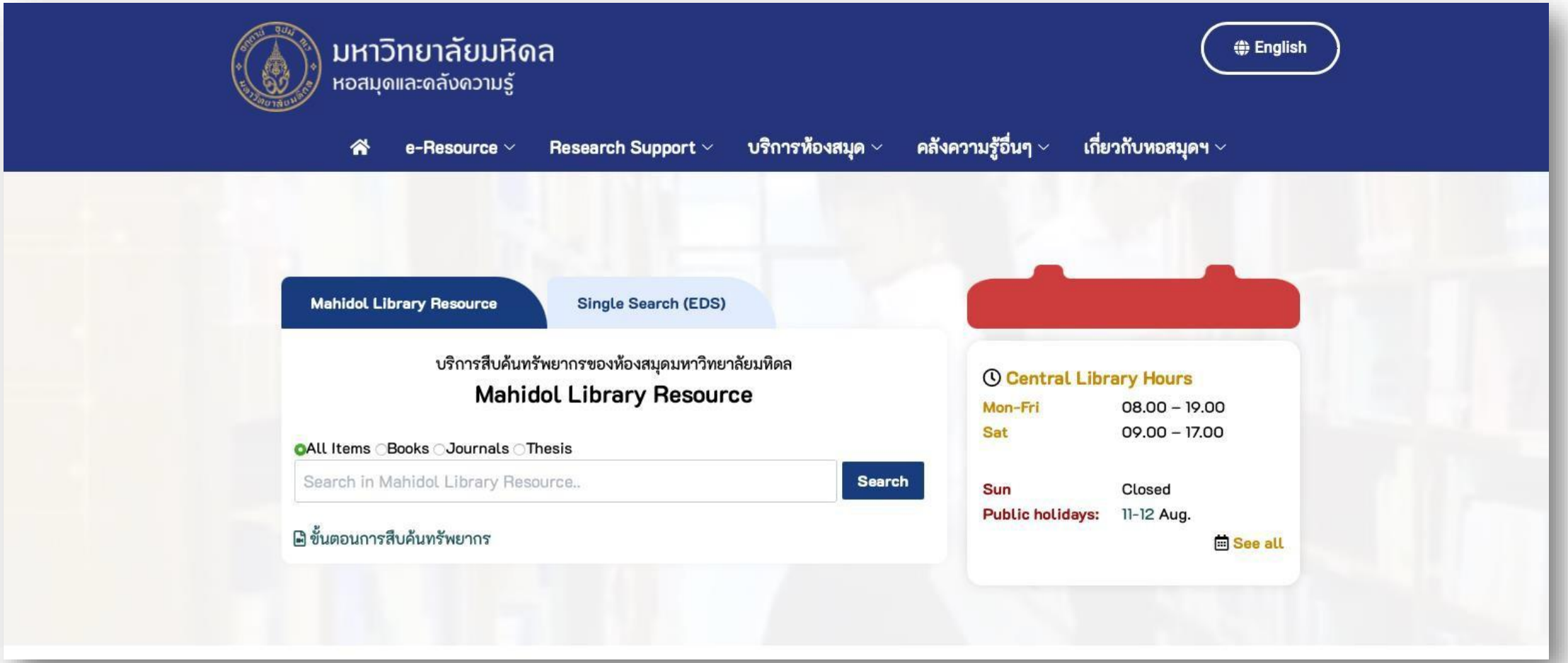
Website with helpful information

- **Library and Knowledge Center**
  http://www.li.mahidol.ac.th



Website with helpful information

- **(1). Library and Knowledge Center http://www.li.mahidol.ac.th**



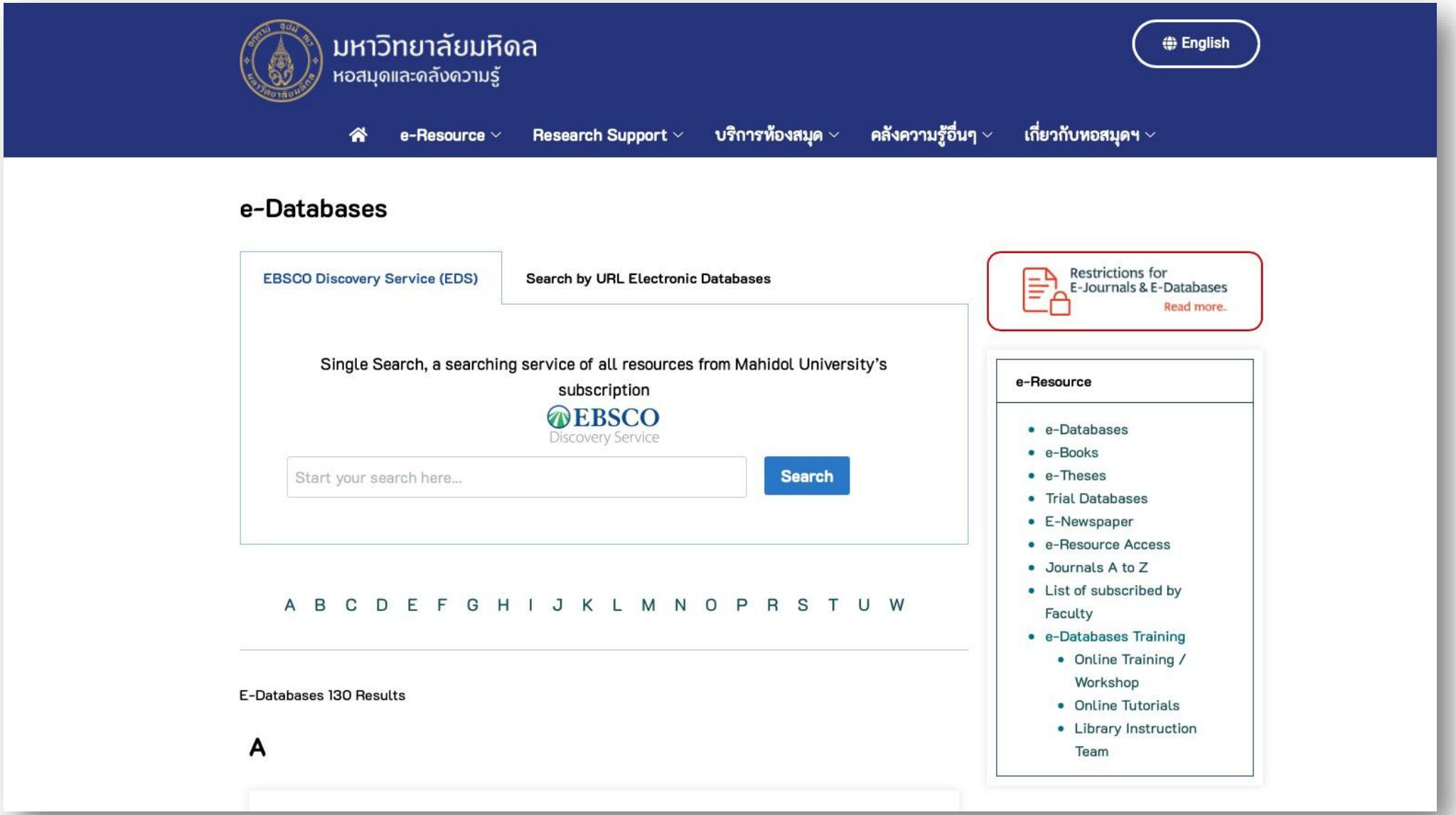
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https://www-li-mahidol-ac-th.ejournal.mahidol.ac.th/e-databases/



MU>= 40 Libs

E-Databases


มหาวิทยาลัยมหิดล
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e-Resource
Research Support
บริการห้องสมุด
คลังความรู้อื่นๆ
เกี่ยวกับหอสมุดฯ
e-Databases
EBSCO Discovery Service (EDS)
Search by URL Electronic Databases
Single Search, a searching service of all resources from Mahidol University's subscription
EBSCO Discovery Service
Start your search here...
Search
A B C D E F G H I J K L M N O P R S T U W
E-Databases 130 Results
A
Restrictions for E-Journals & E-Databases
Read more..
e-Resource
e-Databases
e-Books
e-Theses
Trial Databases
E-Newspaper
e-Resource Access
Journals A to Z
List of subscribed by Faculty
e-Databases Training
Online Training / Workshop
Online Tutorials
Library Instruction Team

Publisher vs Database

- Database
  - Research databases are organized collections of computerized information or data such as periodical articles, books, graphics and multimedia that can be searched to retrieve information.
- Publisher
  - A person or company whose business is the publishing of books, periodicals, computer software, etc.
  - For academic publisher is the subfield of publishing which distributes academic research and scholarship. Most academic work is published in academic journal article, book or thesis form.

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IEEE: Institute of Electrical and Electronics Engineers.

- Founded : January 1, 1963;
- Type: Professional association
- Mission Vision : Objective: IEEE is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity.
- Focus Electrical, Electronics, Communications, Computer Engineering, Computer Science and Information Technology[1]
- Location : Piscataway, New Jersey, US
- Origins: Merger of the American Institute of Electrical Engineers and the Institute of Radio Engineers
- Area served : Worldwide
- Method: Industry standards, Conferences, Publications
- Members: 420,000+
- Key people: James A. Jefferies, President and CEO
- Revenue: US$413 million
- Website: www.ieee.org

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ScienceDirect

- Service by Elsevier.
- Service Type: Database
- Areas: Physical Sciences and Engineering, Life Sciences, Health Sciences, Social Sciences and Humanities
- Number of Document: 16M items
- Online: https://www.sciencedirect.com/
- Can access by using MU network.



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Take a Break!!!



Research Quality

Research Quality

- Peer Review
- Journal Types : Subscription and Open Access Journals
- Impact Factor
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- TCI



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SJR

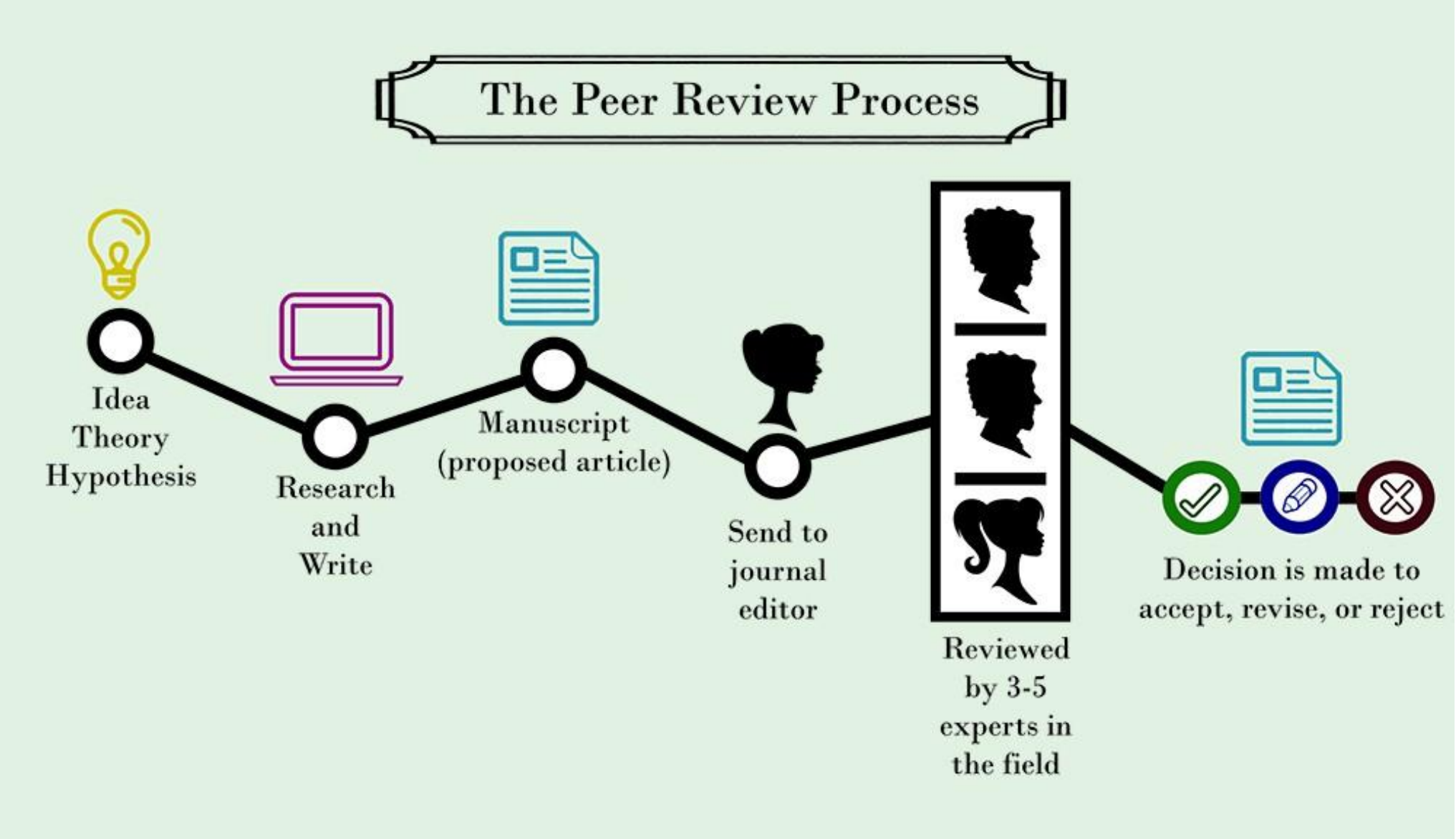


Peer Review

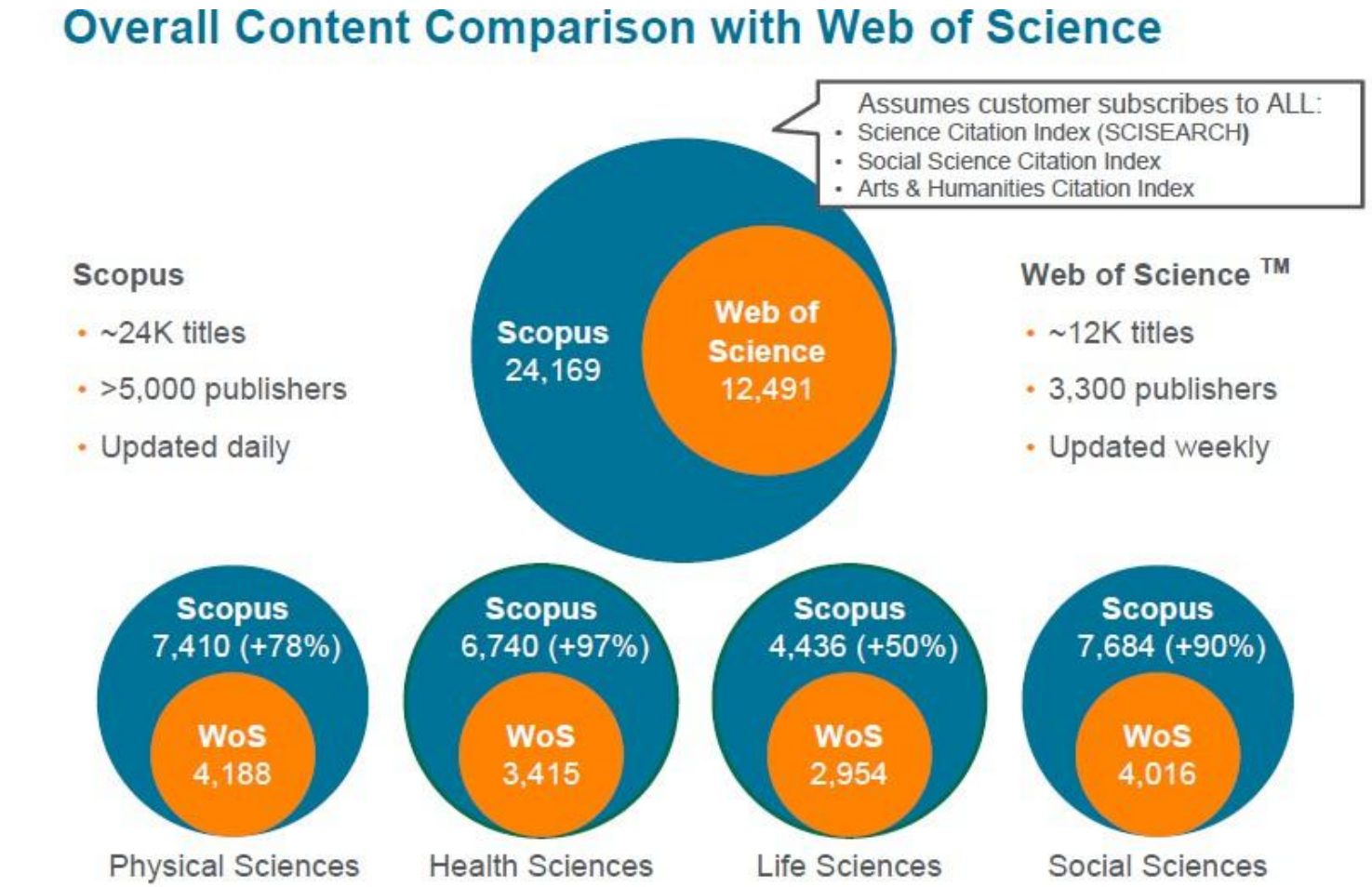
- Peer Review is the evaluation of work by one or more people of similar competence to the producers of the work (peers). It constitutes a form of self-regulation by qualified members of a profession within the relevant field.



Peer Review Process

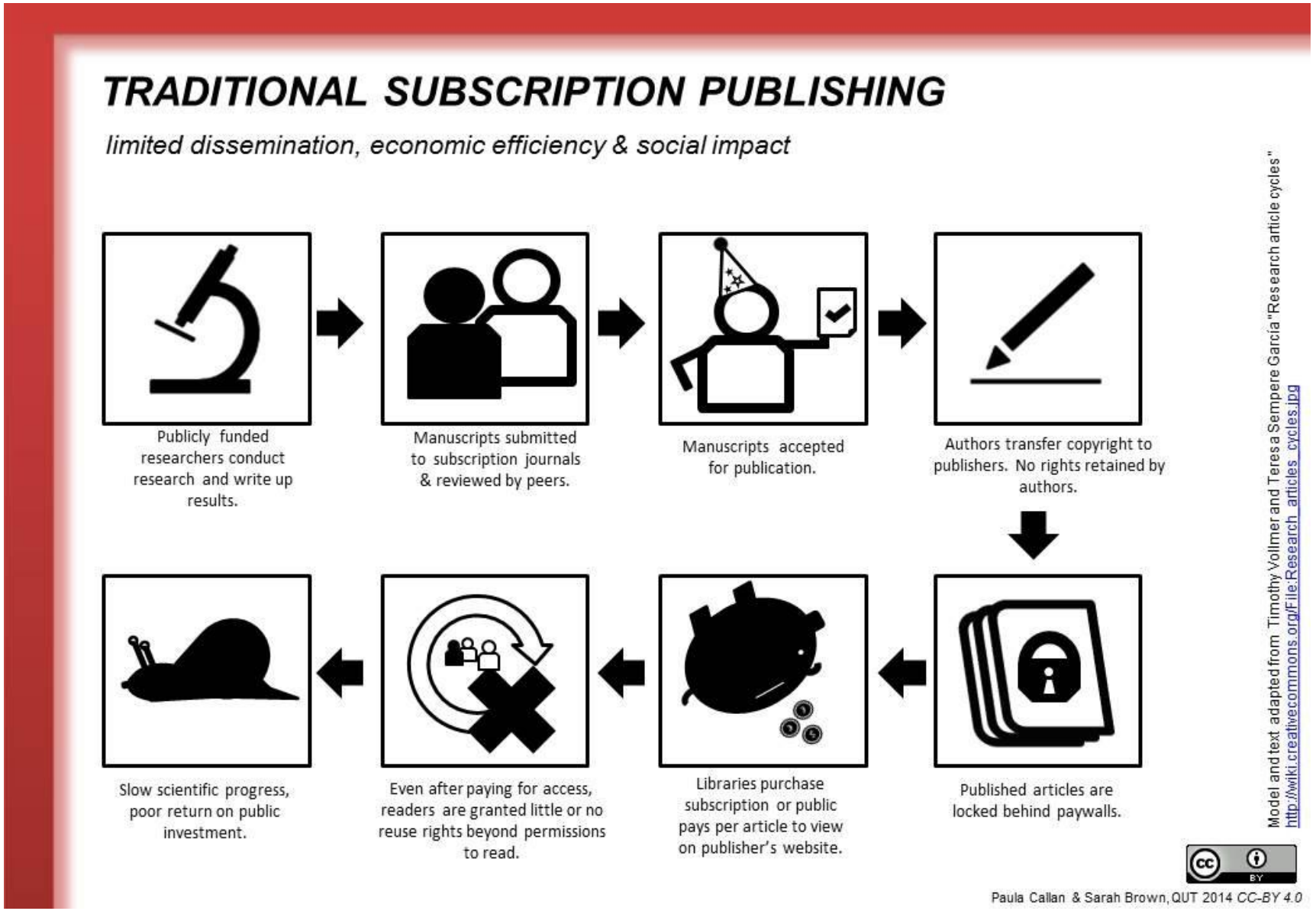


Majority Research Databases

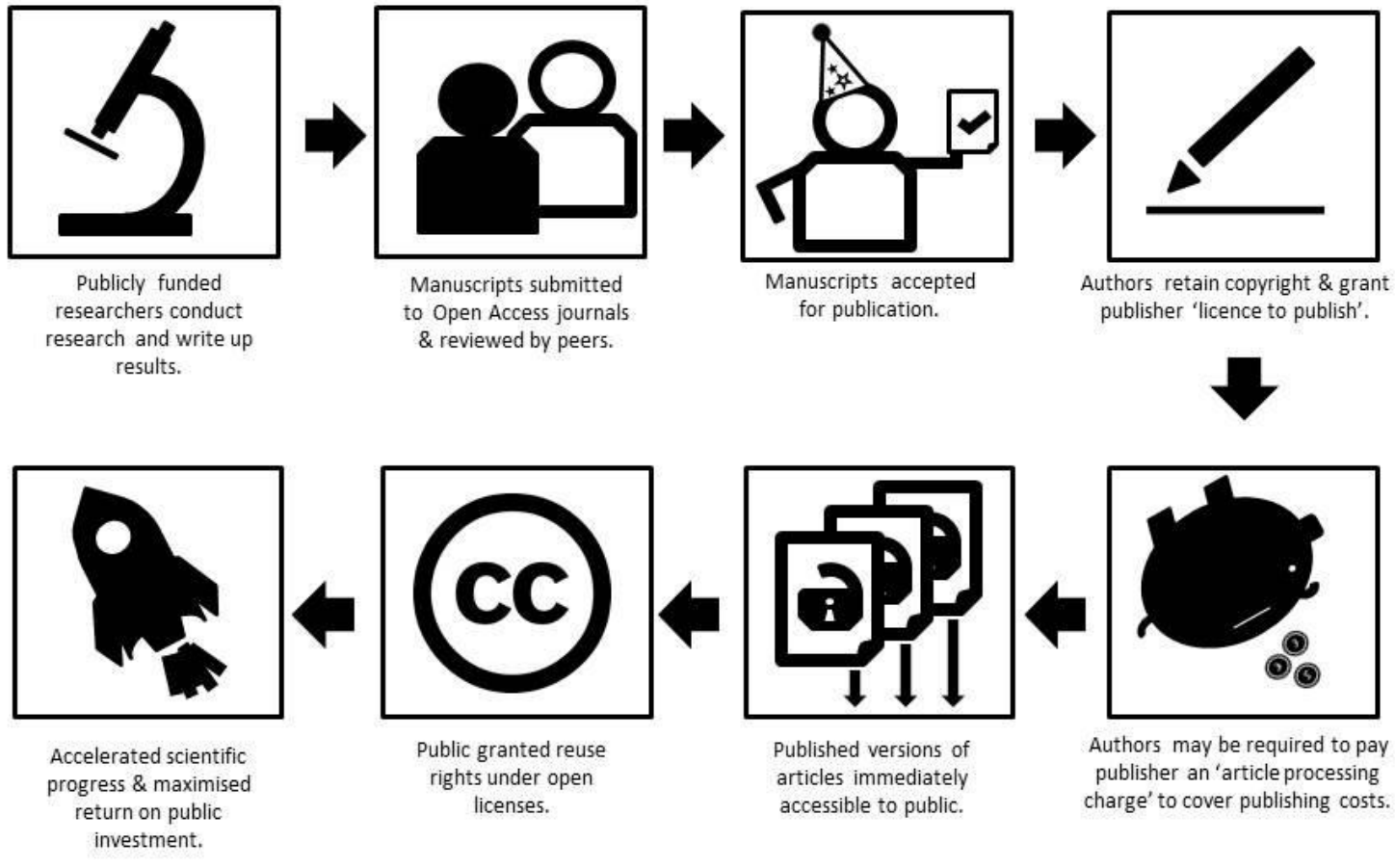


Journal Types: Subscription Vs Open Access

Subscription Journal



Open Access Journal



Comparison of OA and Sub from Elsevier

Open Access	Subscription
Access	
Upon publication articles are immediately and freely available to the wide public.	Articles are made available to subscribers as well as developing countries and patient groups through one of Elsevier's access initiatives.
The final version of articles can immediately be posted to an institutional or subject repository, or to the author's own website. These articles need to bear a CrossMark and link back to the final article on Elsevier's site.	A draft version of the article may be posted voluntarily to own websites or institutional repositories. If posting is mandated, then the mandating organization first needs an agreement with Elsevier.
Fees	
Costs associated with publishing are covered by the Open Access Publication Fee, payable by authors, their institution or their research funder.	Costs associated with publishing are covered by a subscription fee.
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Copyright remains in general with the author(s). See our author agreement for details. The author chooses a Creative Commons license which defines permitted reuse.	Rights are determined by a copyright transfer, where authors retain scholarly rights to post and use their articles, and contracts with subscribing libraries.

Predatory Publishing

Predatory publishing,[3] is an exploitive academic publishing business model that involves charging publication fees to authors without checking articles for quality and legitimacy and without providing the other editorial and publishing services that legitimate academic journals provide, whether open access or not.

Beall's List, a report that was regularly updated by Jeffrey Beall of the University of Colorado until January 2017, set forth criteria for categorizing publications as predatory.[9] Beall took his list offline in January 2017.[10][b]



JCR Impact Factors

- JCR: Journal Citation Reports (JCR) is an annual publication by Clarivate Analytics (previously the intellectual property of Thomson Reuters).
- Impact factor = the ratio of citation (in particular year) to citable items published (in the two previous years)
- คำนวณโดยใช้ข้อมูลการอ้างอิงบทความ จากฐานข้อมูล Institute for Scientific Information (ISI) Web of Knowledge - Web of Science

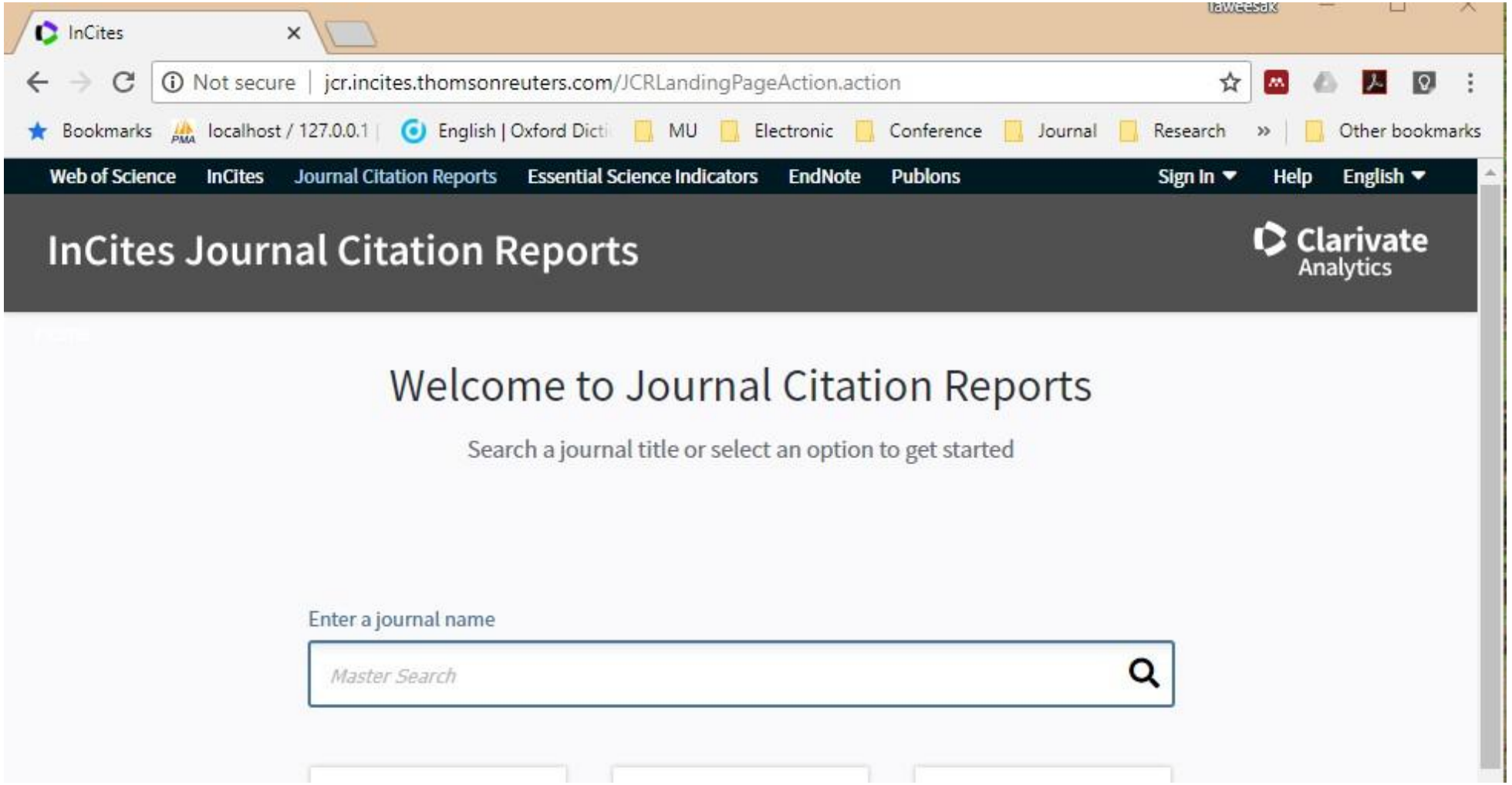
Calculation of JCR Impact factor

- Journal Impact Factor (JIF) = (citations from JCR year to items in "year -2" + citations from JCR year to items in "year -1")/ (citable items in "year -2" + citable items in "year -1") or

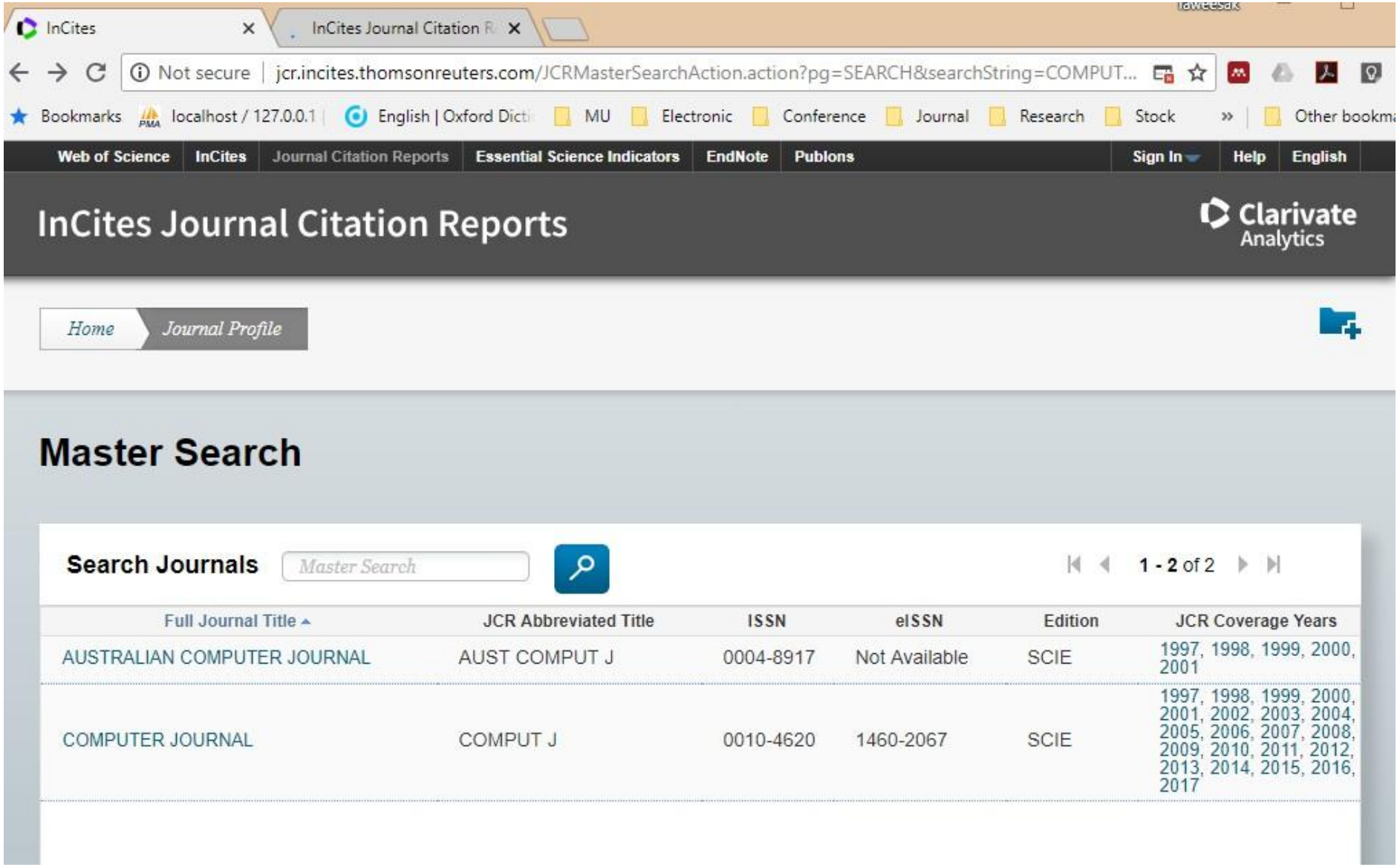
$$IF = \frac{(C_2 + C_1)}{(A_2 + A_1)}$$

where C_1 and C_2 are citations from JCR in year 2 and year 1, A_1 and A_2 are article in this area of year 2 and year 1.

Journal Citation Reports



Example search : “computer”



SJR Indicator

- SJR Indicator is a measure of scientific influence of scholarly journals that accounts for both the number of citations received by a journal and the importance or prestige of the journals where such citations come from.
- A journal's SJR is a numeric value indicating the average number of weighted citations received during a selected year per document published in that journal during the previous three years.
- Higher SJR values are meant to indicate greater journal prestige.

Calculation of SJR indicator

$$SJR_i = \frac{(1-d-e)}{N} + e\frac{Art_i}{\sum_{j=1}^{N} Art_j} + d \cdot \sum_{j=1}^{N} \frac{C_{ji} \cdot SJR_j}{C_j} \cdot \frac{1-\left(\sum_{k \in \{Dangling-nodes\}} SJR_k\right)}{\sum_{h=1}^{N}\sum_{k=1}^{N} \frac{C_{kh} \cdot SJR_k}{C_k}} + d \cdot \left[\sum_{k \in \{Dangling-nodes\}} SJR_k\right] \cdot \frac{Art_i}{\sum_{j=1}^{N} Art_j}$$

$$SJRQ_i = \frac{SJR_i}{Art_i}$$

SJR_i -Scimago Journal Rank of the Journal *i*.

C_{ji} - Citation from journal j to journal *i*.

C_j - Number of References of journal *j*.

d - Constant, normally 0.85.

e - Constant, normally 0.10.

N - Number of Journals

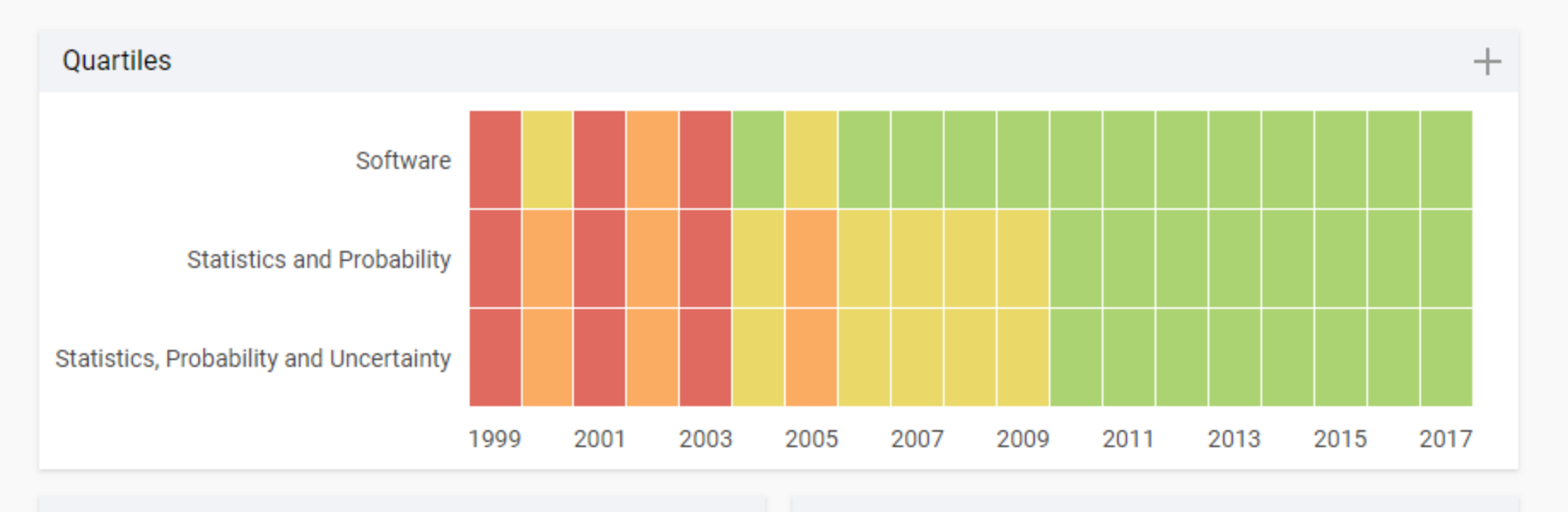
Art_j - Number of Articles of journal *j*

Journal quartile score (Q)

- ค่า Q หมายถึง Quartile score ของวารสารในแต่ละสาขาวิชา (subject categories)
- Q1 = top position (the highest 25% of data)
- เป็นกลุ่มวารสารที่ดีที่สุดในสาขานี้
- Q2 = middle-high position (อยู่ระหว่าง top 50% และ top 25%)
- Q3 = middle-low position (อยู่ อยู่ระหว่าง top 75% และ top 50%)
- Q4 = bottom position (bottom 25%)

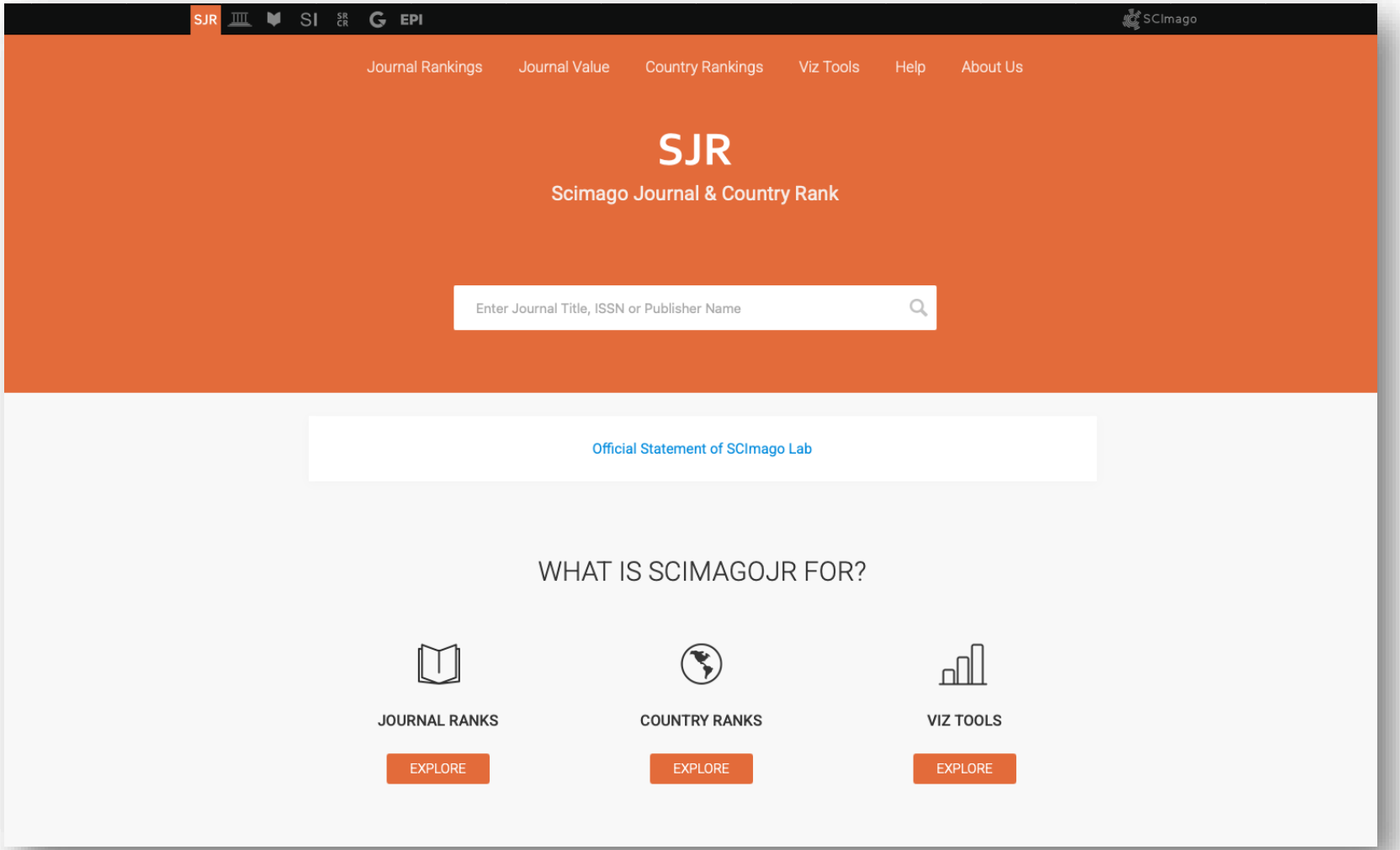
Example of Journal quartile score

- **Journal of Statistical Software**

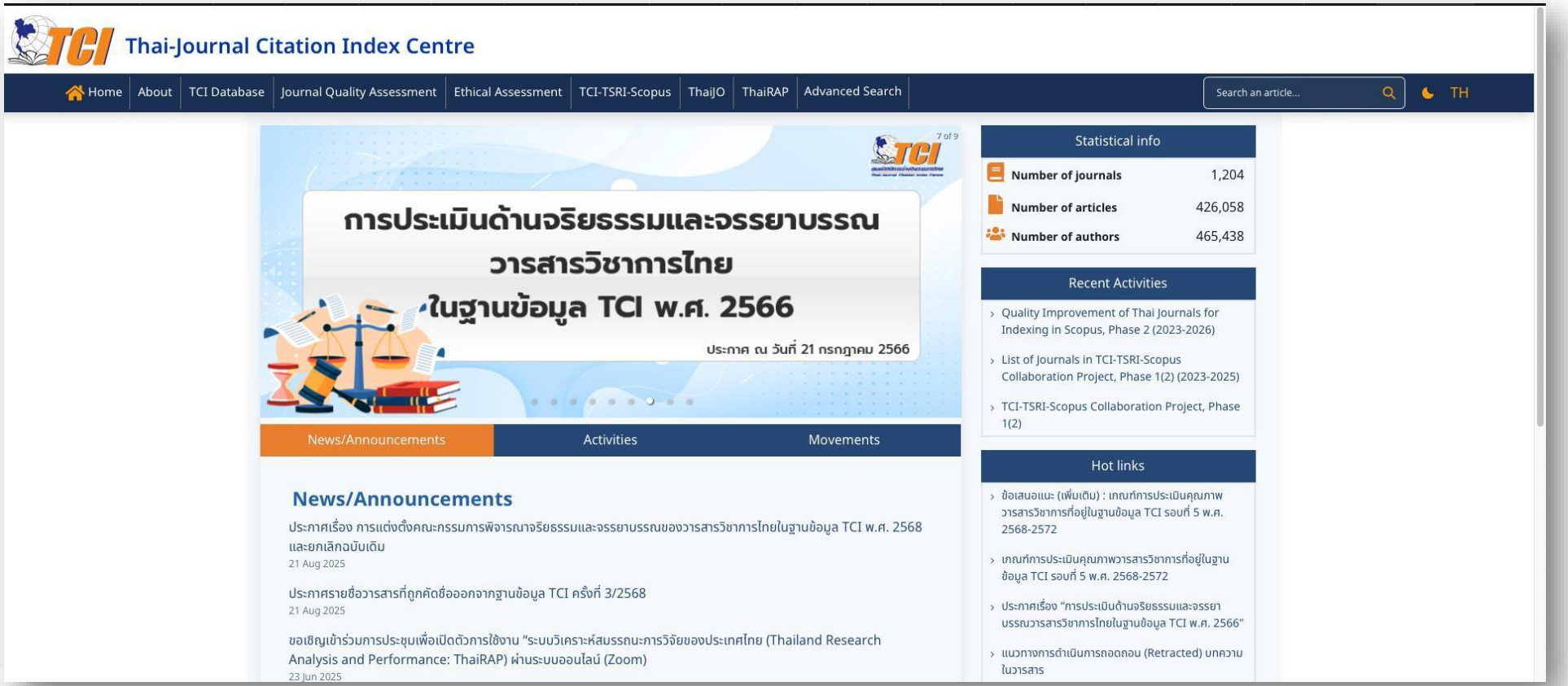


SJR and Journal quartile score Search

- URL: https://www.scimagojr.com

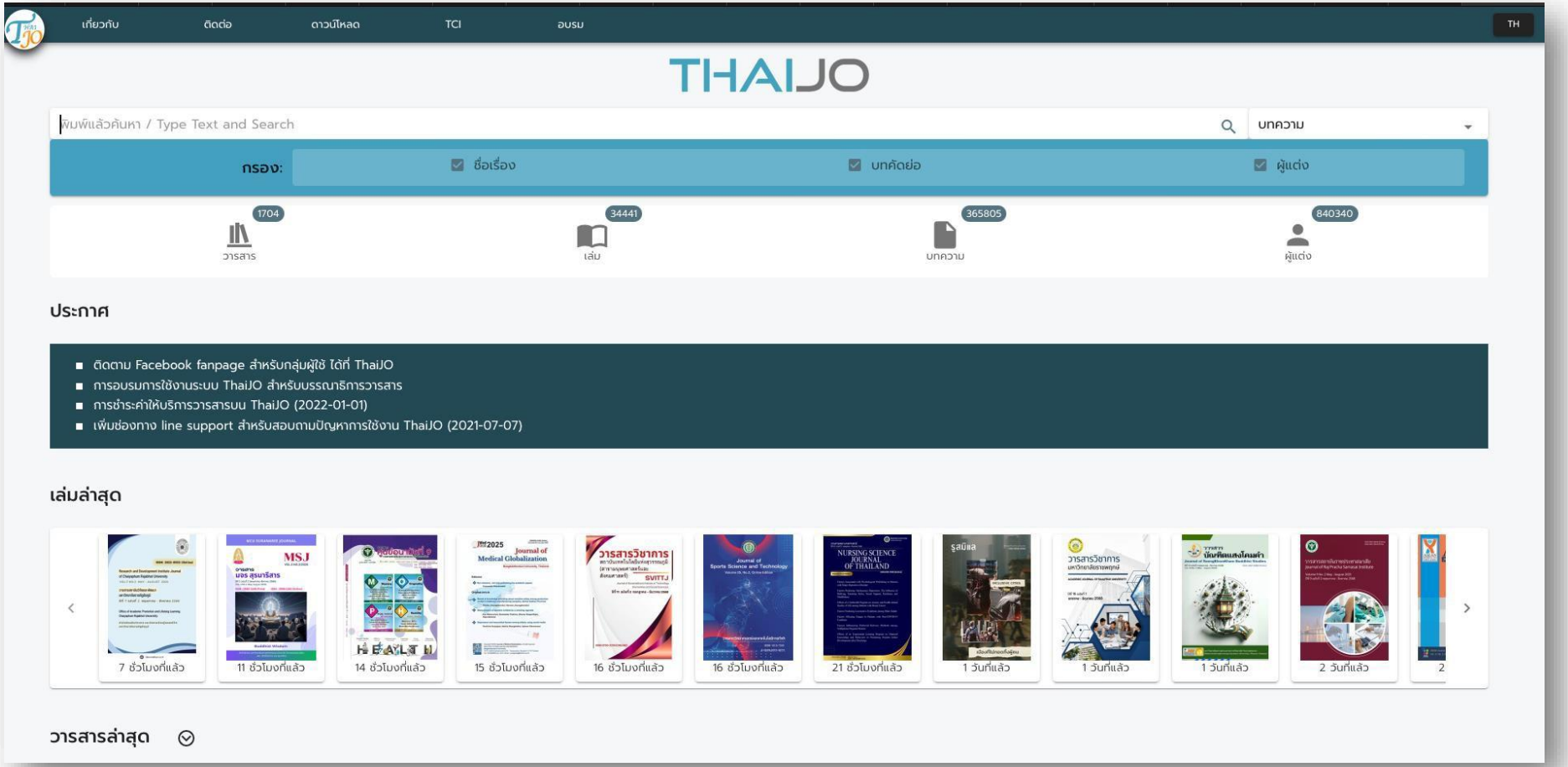


TCI: Thai-Journal Citation Index Centre



https://tci-thailand.org

ThaiJo



https://www.tci-thaijo.org/

The articles for the class activity.

1. Sirichai Panapongpaisan and **Yod Sukamongkol**, “Factors Affecting Human Resource Development for Industry 5.0 in Thailand.” Journal of Mobile Multimedia, Vol 22, Iss1, March 2026
2. Pornphimol Winyuchakrit, Piti Pita, **Yod Sukamongkol**, and Bundit Limmeechokchai, “An Investigation of Internationally Transferred Mitigation Outcomes (ITMOs) on GHG Emissions Reduction in Thailand’s NDC”, *International Energy Journal*, Vol. 25, June 2025
3. Adisorn Leelasantitham, Thammavich Wongsamerchue and **Yod Sukamongkol**, “Economic Pricing in Peer-to-Peer Electrical Trading for a Sustainable Electricity Supply Chain Industry in Thailand”, *Energies*, Vol. 17, No.5, 2024**,** doi.org/10.3390/en17051220
4. Pramote Jaruwatanachai, **Yod Sukamongkol** and Taweesak Samanchuen, “Predicting and Managing EV Charging Demand on Electrical Grids: A Simulation-Based Approach”, *Energies*, Vol 16, 2023, doi.org/10.3390/en16083562
5. Pornphimol Winyuchakrit, **Yod Suakmongkol** and Bundit Limmeechockchai, “Do Electric Vehicles Really Reduce GHG Emissions in Thailand?”, *Energy Procedia*, 2017
6. **Y. Sukamongkol**, S. Chungpaibulpatana, B. Limeechokchai and P. Sripadungtham, “Condenser Heat Recovery with a PV/T Air Heating Collector to Regenerate Desiccant for Reducing Energy Use of an Air Conditioning Room,” *Energy & Buildings*, Vol 42, pp. 315-325, 2010

Group Activity #2

- Selected an article from above, Explore the following information
  - Publisher ?
  - Type of Journal (Subscription / open access)
  - JCR Impact factor?
  - SJR Indicator, Quartile?



Home Work !!

ให้ นศ. ไปศึกษาการใช้งาน VPN เข้าสู่ระบบของมหาวิทยาลัย
แล้วทดลองใช้ฐานข้อมูล E-Database เช่น IEEE xplore, Science Direct
ค้นหาข้อมูลจากภายนอกมหาวิทยาลัย

VPN https://muit.mahidol.ac.th/docs/users-guide/manual-network-and-connectivity/manual-vpn/



Group Assignment Homework #1

- Select your three favorite lecturers in ITM Division, then pick a manuscript for each lecturer and down load.
- From the selected articles, Explore the following information
  - Title of Article?
  - Authors?
  - Title of Journal ?
  - Publisher ?
  - Type of Journal (Subscription / open access)
  - JCR Impact factor?
  - SJR Indicator, Quartile?

Sent this assignment in PDF in LMS